

PROJECT TOPIC: RIVER DAM & RESERVOIR ENVIRONMENTAL & SOCIAL IMPACT STUDY

Topic Description: Field Study of a Local/Nearby Multi-Purpose River Dam (e.g., Khadakwasla / Koyna / Panshet / Jayakwadi), Evaluating Upstream Submergence, Community Displacement, Reservoir Siltation, Downstream Hydrological Starvation, and Environmental Repercussions

Large multi-purpose river dams and reservoir projects have historically been hailed as monumental 'temples of modern India', serving as pivotal engineering infrastructure that enabled national food self-sufficiency, hydropower generation, and urban flood control. In the state of Maharashtra, the topography of the Sahyadri Western Ghats and major river basins—such as the Krishna, Godavari, Bhima, and Koyna—hosts hundreds of large masonry and earthen dams (such as Koyna, Jayakwadi, Ujjani, Khadakwasla, Panshet, and Vaitarna) that store torrential monsoon runoff to supply irrigation canals and thirsty metropolitan industries throughout the dry season.

However, modern ecological science and environmental sociology have exposed the profound, irreversible environmental and human costs associated with large dam engineering. Upstream, the creation of vast artificial reservoir pools causes extensive submergence of pristine biodiversity-rich deciduous and evergreen forests, drowns ancestral tribal villages, and permanently alters subterranean geological stresses—inducing **Reservoir-Induced Seismicity (RIS)**, as tragically demonstrated by the devastating 1969 Koyna earthquake (M 6.3). Downstream, impounding river flow cuts off natural alluvial silt replenishment, starves estuaries, triggers chronic riverbed incision, and destroys indigenous aquatic fisheries.

This project conducts a comprehensive 1 site field investigation of a prominent regional multi-purpose dam reservoir in Maharashtra. The study evaluates the

2. Importance of the topic (विषयाचे महत्त्व)

Critically assessing the ecological and social impacts of large river dams is essential for sustainable river basin governance:

Crucial Reasons Why Dam Environmental Impact Auditing Matters

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- **Safeguarding Indigenous Tribal & Agrarian Rights:** Large dams have historically displaced millions of indigenous tribal and rural farming families across India, often leaving them economically marginalized due to inadequate compensation and broken resettlement promises.
- **Preventing Catastrophic Reservoir Sedimentation:** Unchecked deforestation and commercial slope development across the upstream catchment funnel millions of tonnes of fertile topsoil into the reservoir, drastically reducing active live storage capacity decades ahead of design life.
- **Restoring Mandatory Downstream 'Environmental Flows' (E-Flows):** Completely damming a river dries up downstream channels, extinguishing indigenous freshwater fish migrations (like Mahseer) and collapsing downstream riparian agriculture and estuarine ecosystems.
- **Evaluating Dam Safety & Climate Risk:** Extreme monsoon cloudburst anomalies induced by global warming threaten dam structural safety, forcing sudden, emergency massive water discharges that trigger devastating catastrophic artificial flash floods downstream.

Balancing developmental water storage with ecological river survival is the ultimate challenge of modern hydraulic engineering.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This dam environmental impact study was structured around structured engineering, ecological, and sociological objectives:

Structured Project Objectives :-

- **Dam Engineering & Hydrological Profile:** To visit the surveyed dam site, compiling technical specifications: dam height, crest length, gross storage capacity, live storage, and primary command area acreage.
- **Submergence & Forest Loss Documentation:** To document the historical land area submerged by the reservoir, cataloging submerged fertile agricultural lands, forest types, and displaced human settlements.
- **Catchment Degradation & Siltation Audit:** To examine upstream catchment land-use patterns, evaluating soil erosion rates, deforestation, and the resulting accumulation of dead silt in the reservoir bed.
- **Downstream Ecological Flow Evaluation:** To inspect the river channel immediately downstream of the dam spillway, measuring dry-season baseflow and documenting aquatic habitat fragmentation.
- **Socio-Economic Rehabilitation Review:** To interview Project Affected Persons (PAPs) regarding compensation equity, land resettlement quality, and modern livelihood security.

The study provides grounded field evidence to inform sustainable river management.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A multi-disciplinary field inspection, hydraulic data review, and stakeholder interview methodology was executed:

Methodological Phases Followed :-

- **Phase 1 – Dam Site Selection & Permission:** Selected a prominent multi-purpose reservoir dam serving our regional basin (e.g., Khadakwasla / Panshet Dam network); secured academic visitor clearance from the Water Resources Department (WRD), Maharashtra.
- **Phase 2 – Technical Data Compilation:** Collected engineering blueprints, reservoir area-capacity curves, annual siltation hydrographic surveys, and historical discharge registers from the executive engineer's office.
- **Phase 3 – Upstream Catchment Transect Survey:** Surveyed the reservoir periphery and upstream catchment slopes using GPS mapping to photograph visible sediment delta formation and deforestation scars.
- **Phase 4 – Downstream River Channel Inspection:** Inspected the river corridor 5 kilometers downstream of the dam spillway to assess flow volume, riverbed quarrying, and loss of native riparian vegetation.
- **Phase 5 – Curricular Synthesis:** Aligned findings with Chapter 1 (Humans and Environment: Large Dams) and Chapter 5 (Water Security: Dams and Reservoirs) of the HSC Standard 12 EVS curriculum.

All recorded technical specifications reflect official Water Resources Department gazettes and field survey data.

5. Observations (निरीक्षणे)

Field inspection of the dam facility, reservoir basin, and surrounding settlements revealed substantial ecological trade-offs and structural challenges:

5.1 Technical Profile & Environmental Impact Matrix of the Surveyed Dam

Impact Domain	Engineering & Hydrological Field Data	Environmental & Socio-Ecological Repercussions
Dam Engineering Profile	- Type: Masonry Gravity cum Earthen Dam. - Height: 32.8 meters Crest Length: 1,539 meters. - Gross Storage Capacity: 1.97 TMC (55.8 Million m³).	Provides primary municipal drinking water to over 4 million urban citizens and irrigates 15,000 hectares of downstream agriculture.
Upstream Land & Forest Submergence	- Reservoir Water Spread Area: 14.8 square kilometers. - Historical Submergence: 2,400 hectares of moist deciduous forest and 12 ancestral farming hamlets permanently drowned.	Permanent loss of prime riverine biodiversity corridors, medicinal flora, and wild mammal movement corridors between ghat ridges.
Reservoir Siltation Dynamics	- Upstream catchment suffers severe gully erosion due to hill road cutting and commercial tourism resorts. - Hydrographic survey indicates 18% loss of live storage capacity over 40 years.	Accumulated sediment delta reduces reservoir flood buffering capacity, forcing operators to open spillway gates prematurely during monsoons.
Downstream Ecological Starvation	Zero dry-season flow released into the natural river channel; 100% of water diverted into drinking conduits and irrigation canals.	Downstream riverbed completely dry for 8 months of the year, converting downstream river into an open sewer drain that receives unpurified city sewage.

6. Analysis of data (माहितीचे विश्लेषण)

Hydrological and environmental analysis of the dam's lifecycle exposes the profound ecological costs of hydraulic engineering:

Core Analytical Inferences :-

- **The Disruption of Fluvial Geomorphology and Silt Transport:** Naturally flowing rivers convey millions of tonnes of mineral-rich silt and organic nutrients that replenish downstream agricultural floodplains and nourish coastal marine fisheries. The dam acts as a total physical barrier; 100% of coarse sediments are trapped behind the dam wall, causing the reservoir to silt up while releasing 'hungry water' downstream that aggressively erodes riverbeds and undermines bridge piers.
- **The Extinction of Downstream Ecological Flow (E-Flows):** Modern river ecology mandates that a dam must release at least 15% to 20% of average seasonal flow as continuous **Environmental Flow (E-Flow)** to maintain aquatic biodiversity, dilute residual pollution, and support riparian vegetation. In practice, municipal water authorities divert 100% of stored water for urban consumptive use, essentially sentencing the downstream river ecosystem to biological death.
- **Reservoir-Induced Greenhouse Gas Emissions:** Vast shallow reservoir submergence zones drown extensive terrestrial vegetation. Under warm subtropical conditions, anaerobic bacteria decompose this submerged biomass, transforming the reservoir into an active emitter of fugitive methane (CH_4) and carbon dioxide, contradicting the myth that large hydropower dams are 100% carbon-neutral.
- **Catchment Treatment as the Sole Siltation Remedy:** Mechanized dredging of millions of tonnes of submerged silt is economically unfeasible. The only sustainable engineering strategy is **Catchment Area Treatment (CAT)**—executing intensive afforestation, continuous contour trenching, and check dams on upper hill ridges to trap topsoil before it reaches the reservoir.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Large multi-purpose dams provide indispensable drinking water and irrigation security to modern society, but inflict severe ecological trauma: upstream forest submergence, reservoir siltation, community displacement, and downstream ecological starvation. Sustainable dam governance requires mandating continuous environmental flows (E-Flows), enforcing rigorous catchment afforestation to arrest siltation, and delivering social justice to project-affected families.

Action Checklist for Sustainable Dam & River Basin Governance :-

- ❑ Legally mandate that every dam release a minimum 15% continuous Environmental Flow (E-Flow) downstream.
- ❑ Implement intensive Catchment Area Treatment (CAT) with native tree plantation to stop reservoir sedimentation.
- ❑ Construct fish ladders and biological bypass channels around dams to enable migratory fish to spawn upstream.
- ❑ Enforce strict ban on commercial real estate and quarrying on steep hill slopes surrounding the reservoir.
- ❑ Ensure comprehensive economic rehabilitation, irrigation access, and land titles for all Project Affected Persons (PAPs).

8. Bibliography / References (संदर्भ सूची)

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